

Python Tuple - Complete Documentation

For BBA Apex Students

1. Introduction

A tuple is an ordered, immutable collection of elements in Python.

Key Features:

- Ordered
- Immutable
- Allows duplicates
- Can store multiple data types

```
In [20]: # Creating a tuple
t = (1, 2, 3)
type(t)
# print(t)
```

Out[20]: tuple

```
In [57]: a=(1,2,3,4,5)
b=(6,7,8,9,10)

2 not in a
```

Out[57]: False

```
In [30]: a=[1,2,3,4,5]
qweee=tuple(a)
type(qweee)
```

Out[30]: tuple

```
In [48]: t=(10,20,30,40,50)
t[3:]
```

Out[48]: (40, 50)

2. Creating Tuples

- Basic tuple
- Single element tuple
- Empty tuple
- Using tuple() constructor

```
In [10]: # Single element tuple
t1 = (5,)
```

```
print(type(t1))

# Empty tuple
t2 = ()
print(t2)

# Using constructor
t3 = tuple([1, 2, 3])
print(t3)
```

```
<class 'tuple'>
()
(1, 2, 3)
```

3. Accessing Tuple Elements

- Positive indexing
- Negative indexing
- Slicing
- Nested access

```
In [52]: t = (10, 20, 30, 40, 50)
t.append(100)
print(t)

# Indexing
# print(t[0])
# print(t[-1])

# # Slicing
# print(t[1:4])

# Nested tuple
# nested = ((1,2),(3,4))
# print(nested[1][0])
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[52], line 2
      1 t = (10, 20, 30, 40, 50)
----> 2 t.append(100)
      3 print(t)
      5 # Indexing
      6 # print(t[0])
      7 # print(t[-1])
      (...)      13 # nested = ((1,2),(3,4))
      14 # print(nested[1][0])

AttributeError: 'tuple' object has no attribute 'append'
```

4. Immutability

Tuples cannot be modified after creation.

```
In [4]: t = (1,2,3)
        #t[0] = 10 # This will raise TypeError
```

```
-----
-
TypeError                                Traceback (most recent call las
t)
Cell In[4], line 2
      1 t = (1,2,3)
----> 2 t[0] = 10 # This will raise TypeError

TypeError: 'tuple' object does not support item assignment
```

5. Tuple Operations

- Concatenation
- Repetition
- Membership

```
In [2]: t1 = (1,2)
        t2 = (3,4)

        print(t1 + t2)
        print(t1 * 2)
        print(2 in t1)
```

```
(1, 2, 3, 4)
(1, 2, 1, 2)
True
```

6. Built-in Functions

- len()
- max()
- min()
- sum()

```
In [58]: t = (5,10,15)
          print(len(t))
          print(max(t))
          print(min(t))
          print(sum(t))
```

```
3
15
5
30
```

7. Tuple Methods

- count()
- index()

```
In [12]: t = (1,2,2,3)
          print(t.count(2))
```

```
print(t.index(3))
```

2
3

8. Packing and Unpacking

```
In [13]: t = 10,20,30 # Packing  
a,b,c = t # Unpacking  
print(a,b,c)
```

10 20 30

```
In [15]: for i in range(2,10,1):  
        print(i*2)
```

4
6
8
10
12
14
16
18

```
In [61]: a=(1,2)  
b,d=a
```

```
In [62]: d
```

Out[62]: 2

```
In [63]: b
```

Out[63]: 1

```
In [ ]:
```